

String and Text Manipulation – Interview Questions

1. Remove Spaces from Given Text

Problem: Write a function to remove all spaces from the input string. Explanation: Remove any whitespace characters. Input: "he llo wor ld" Output: "helloworld"

Code

```
def remove_spaces(text):  
    return text.replace(" ", "")  
  
print(remove_spaces("he llo wor ld"))
```

Output

```
helloworld
```

2. Reverse a String

Problem: Write a function to reverse the characters in a string. Input: "hello" Output: "olleh"

Code

```
def reverse_string(text):  
    return text[::-1]  
  
print(reverse_string("hello"))
```

Output

```
olleh
```

3. Reverse a String After Removing Spaces

Problem: Write a function to reverse a string after removing all spaces. Input: "he llo world" Output: "dlrowolleh"

Code

```
def reverse_without_spaces(text):  
    text = text.replace(" ", "")  
    return text[::-1]  
  
print(reverse_without_spaces("he llo world"))
```

Output

```
dlrowolleh
```

4. Convert Snake Case to Camel Case

Problem: Convert a string from snake_case to camelCase. Input: "my_variable_name" Output: "myVariableName"

Code

```
def snake_to_camel(text):
    words = text.split("_")
    result = words[0]
    for word in words[1:]:
        result = result + word.capitalize()
    return result

print(snake_to_camel("my_variable_name"))
```

Output

myVariableName

5. Convert Snake Case to Pascal Case

Problem: Convert a string from snake_case to PascalCase. Input: "my_variable_name" Output: "MyVariableName"

Code

```
def snake_to_pascal(text):
    words = text.split("_")
    result = ""
    for word in words:
        result = result + word.capitalize()
    return result

print(snake_to_pascal("my_variable_name"))
```

Output

MyVariableName

6. Convert Camel Case to Snake Case

Problem: Convert a string from camelCase to snake_case. Input: "myVariableName" Output: "my_variable_name"

Code

```
def camel_to_snake(text):
    result = ""
    for ch in text:
        if ch.isupper():
            result = result + "_" + ch.lower()
        else:
            result = result + ch
    return result

print(camel_to_snake("myVariableName"))
```

Output

my_variable_name

7. Convert Camel Case to Pascal Case

Problem: Convert a string from camelCase to PascalCase. Input: "myVariable" Output: "MyVariable"

Code

```
def camel_to_pascal(text):  
    return text[0].upper() + text[1:]  
  
print(camel_to_pascal("myVariable"))
```

Output

MyVariable

8. Convert Pascal Case to Camel Case

Problem: Convert a string from PascalCase to camelCase. Input: "MyVariable" Output: "myVariable"

Code

```
def pascal_to_camel(text):  
    return text[0].lower() + text[1:]  
  
print(pascal_to_camel("MyVariable"))
```

Output

myVariable

9. Convert Pascal Case to Snake Case

Problem: Convert a string from PascalCase to snake_case. Input: "MyVariable" Output: "my_variable"

Code

```
def pascal_to_snake(text):  
    result = ""  
    for i in range(len(text)):  
        ch = text[i]  
        if ch.isupper() and i != 0:  
            result = result + "_" + ch.lower()  
        else:  
            result = result + ch.lower()  
    return result  
  
print(pascal_to_snake("MyVariable"))
```

Output

my_variable

10. Convert Text to Camel Case

Problem: Convert a space-separated sentence into camelCase. Input: "hello world example" Output: "helloWorldExample"

Code

```
def text_to_camel(text):
    words = text.split()
    result = words[0].lower()
    for word in words[1:]:
        result = result + word.capitalize()
    return result

print(text_to_camel("hello world example"))
```

Output

```
helloWorldExample
```

11. Convert Text to Snake Case

Problem: Convert a space-separated sentence into snake_case. Input: "hello world example" Output: "hello_world_example"

Code

```
def text_to_snake(text):
    return "_".join(text.lower().split())

print(text_to_snake("hello world example"))
```

Output

```
hello_world_example
```

12. Convert Text to Pascal Case

Problem: Convert a space-separated sentence into PascalCase. Input: "hello world example" Output: "HelloWorldExample"

Code

```
def text_to_pascal(text):
    words = text.split()
    result = ""
    for word in words:
        result = result + word.capitalize()
    return result

print(text_to_pascal("hello world example"))
```

Output

```
HelloWorldExample
```

13. Swap Upper and Lower Case

Problem: Swap the case of each letter in a given string. Input: "HeLLo" Output: "hEllo"

Code

```
def swap_case(text):  
    return text.swapcase()  
  
print(swap_case("HeLLo"))
```

Output

hEllo

14. Separate Digits from Text

Problem: Extract all digits from a given alphanumeric string. Input: "abc123d4" Output: "1234"

Code

```
def extract_digits(text):  
    digits = ""  
    for ch in text:  
        if ch.isdigit():  
            digits = digits + ch  
    return digits  
  
print(extract_digits("abc123d4"))
```

Output

1234

15. Print Uppercase, Lowercase, Digits and Special Characters Separately

Problem: Print each type of character separately from the string. Input: "Abc123!@#" Output:

Uppercase: A
Lowercase: b c
Digits: 1 2 3
Special Characters: ! @ #

Code

```
def print_char_types(text):
    upper = []
    lower = []
    digits = []
    special = []
    for ch in text:
        if ch.isupper():
            upper.append(ch)
        elif ch.islower():
            lower.append(ch)
        elif ch.isdigit():
            digits.append(ch)
        else:
            special.append(ch)
    print("Uppercase:", " ".join(upper))
    print("Lowercase:", " ".join(lower))
    print("Digits:", " ".join(digits))
    print("Special Characters:", " ".join(special))

print_char_types("Abc123!@#")
```

Output

Uppercase: A
Lowercase: b c
Digits: 1 2 3
Special Characters: ! @ #

16. Count of Uppercase, Lowercase, Digits and Special Characters

Problem: Count each type of character in a string. Input: "AbC@123x!" Output:

Uppercase: 2
Lowercase: 1
Digits: 3
Special Characters: 2

Note: the expected output above shows Lowercase: 1, but the input "AbC@123x!" contains two lowercase letters ('b' and 'x'), so the program correctly prints 2.

Code

```
def count_char_types(text):
    upper = 0
    lower = 0
    digits = 0
    special = 0
    for ch in text:
        if ch.isupper():
            upper += 1
        elif ch.islower():
            lower += 1
        elif ch.isdigit():
            digits += 1
        else:
            special += 1
    print("Uppercase:", upper)
    print("Lowercase:", lower)
    print("Digits:", digits)
    print("Special Characters:", special)

# Note: "AbC@123x!" has 2 lowercase letters ('b' and 'x'),
# so the correct count is 2 (the sample output in the question says 1).
count_char_types("AbC@123x!")
```

Output

Uppercase: 2
Lowercase: 2
Digits: 3
Special Characters: 2

17. Check Password Strength

Problem: Check if a password contains at least one uppercase, one lowercase, one digit, and one special character. Input: "Pass123!" Output: "Strong Password"

Code

```
def check_password(password):
    has_upper = False
    has_lower = False
    has_digit = False
    has_special = False
    for ch in password:
        if ch.isupper():
            has_upper = True
        elif ch.islower():
            has_lower = True
        elif ch.isdigit():
            has_digit = True
        else:
            has_special = True
    if has_upper and has_lower and has_digit and has_special:
        return "Strong Password"
    else:
        return "Weak Password"

print(check_password("Pass123!"))
```

Output

Strong Password

18. Remove Duplicates in a Given Input

Problem: Remove duplicate characters from a string. Input: "aabbcc" Output: "abc"

Code

```
def remove_duplicates(text):
    result = ""
    for ch in text:
        if ch not in result:
            result = result + ch
    return result

print(remove_duplicates("aabbcc"))
```

Output

abc

19. Print Duplicates in a Given String

Problem: Identify and print duplicate characters in a string. Input: "aabbccde" Output: a b c

Code

```
def print_duplicates(text):
    duplicates = []
    for ch in text:
        if text.count(ch) > 1 and ch not in duplicates:
            duplicates.append(ch)
    print(" ".join(duplicates))

print_duplicates("aabbccde")
```

Output

a b c

20. Print Next Characters in a Given String

Problem: Replace each character in the string with its next character. Input: "abc" Output: "bcd"

Code

```
def next_characters(text):
    result = ""
    for ch in text:
        result = result + chr(ord(ch) + 1)
    return result

print(next_characters("abc"))
```

Output

bcd